

A Survey of Constraints for the Examination Timetabling Problem

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Abstract

Exam timetabling in higher education involves numerous regulatory and organizational constraints that are often institution-specific. Despite many algorithmic approaches showing promising results on simplified datasets, practical adoption remains limited due to a lack of alignment with real-world needs. This paper presents a comprehensive catalogue of constraints collected from multiple institutions, aiming to improve modularity in scheduling models and facilitate communication between IT teams and stakeholders to ease adoption of automated scheduling solution.

Keywords: Exam timetabling, Scheduling, Constraint programming, Higher education

1 Introduction

Examinations are a central component of academic programs. They generate significant stress for students and place considerable pressure on institutional resources. To mitigate these challenges and comply with stringent regulations, institutions must

047 allocate substantial human resources to construct examination timetables that sat-
048 isfy regulatory requirements, human resource constraints, and student well-being
049 considerations, all within limited material capacities.
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052 With the advent of computers in administration, IT departments and research
053 teams have sought to automate part, if not all, of the examination scheduling process.
054 While numerous algorithms have demonstrated satisfactory performance within the-
055 oretical frameworks and on simplified datasets, practical adoption within institutions
056 remains limited. This gap is in part due to the misalignment between existing soft-
057 ware solutions and the heterogeneous, institution-specific constraints encountered in
058 real-world examination scheduling.
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061 In this paper, we address this issue by proposing a diverse and comprehensive cata-
062 logue of examination scheduling constraints, collected from multiple institutions. This
063 catalogue is designed to promote modular modeling approaches and to facilitate com-
064 munication between IT teams and the various stakeholders involved in examination
065 scheduling.
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067 068 069 070 071 072 073 **2 Related works** 074

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076 Various constraint collections have typically been conducted at the departmental or
077 institutional level, such as [Cowling et al. \(2002\)](#) for the University Technology MARA.
078 However, the most ambitious constraint collection effort was carried out by Edmund
079 Burke, Dave Elliman, Peter Ford, and Rupert Weare in [Burke et al. \(1996\)](#), in which
080 the authors gathered information on constraints and usage related to examination
081 scheduling from ninety-five British universities. More recently [Siew et al. \(2024\)](#) did
082 a survey on solution Methodologies for exam timetabling problems that included a
083 survey of constraints used.
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086 Nonetheless, most sources of constraints in the exam scheduling problem come
087 from articles focused on solution methods, where researchers use constraints from
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their own institutions to build plausible models for evaluating the effectiveness of their approaches, such as [Bergmann and Zurheide \(2014\)](#).

A recent case study by [Chaplin et al. \(2025\)](#) likewise reports that mismatches between institutional practices and off-the-shelf scheduling tools remain a primary barrier to adoption.

3 Methodology

3.1 Constraint Collection

For this survey, we developed three approaches to constraint collection. The first consists of a survey of the administrative teams at Université de Tours. The second is based on the analysis of models used in scientific articles. Finally, we constructed additional constraints by deriving variations from existing ones.

3.1.1 Survey at Université de Tours

The first source of constraints comes from fieldwork conducted at the University of Tours, with a focus on the Faculty of Science and Technology and the central Teaching Directorate (DIFOR). The objective was to collect real, institution-specific constraints encountered in the exam scheduling process.

Interviews were conducted with administrative staff at the Faculty of Science and Technology. These discussions revealed the practical constraints that arise in scheduling examinations across a variety of programmes, including special diplomas and students with specific statuses (e.g., double degrees, part-time students). The resulting list of constraints was reviewed and validated by faculty administrators to ensure it accurately reflects operational requirements.

During the course of the research, discussions were held with staff members from DIFOR, the university’s central teaching directorate. Among its many responsibilities, DIFOR provides support to faculty administrative teams in managing the timetabling

139 software. These interviews uncovered additional constraints not present in the Fac-
140 ulty of Science but commonly encountered in other departments. The inclusion of
141 these broader constraints helped to generalize the catalogue beyond the initial faculty
142 context and ensured wider applicability across institutional settings.
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145 146 **3.1.2 Extraction from Research Literature** 147

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149 To complement the institutional survey, we conducted a structured literature scan to
150 identify additional constraints reported in academic research. Because the examination
151 scheduling problem spans multiple disciplines including computer science, operations
152 research, mathematics, sociology or management, no single journal or bibliographic
153 database provides comprehensive coverage. We initially relied on Google Scholar,
154 using the search terms “exam timetabling” and “exam scheduling”. Publications were
155 reviewed iteratively until saturation was reached, defined as four consecutive papers
156 introducing no new constraints. In most cases, constraints were presented only as
157 contextual elements to evaluate algorithmic performance; nevertheless, any explicitly
158 stated or implicitly required constraint was extracted and incorporated into the cat-
159 alogue. We did this procedure until the 7th July 2025. To ensure broader and more
160 systematic coverage, the 15th of November 2025, we manually examined all papers con-
161 taining the keyword “exam” from the two major conferences dedicated to automated
162 timetabling and scheduling, namely PATAT and MISTA, as well as from the Jour-
163 nal of Scheduling. In addition, publisher portals were used to retrieve relevant articles
164 from a wide range of journal available in Appendix A. We also reviewed conference
165 proceedings from CPAIOR, ICAPS (13th to 34th editions), the European Conference
166 on Planning, the Conference on Artificial Intelligence Planning Systems (excluding the
167 first edition), and the fourth and fifth editions of the European Conference on Plan-
168 ning, as well as from the Expert Planning Systems (1990) workshop. This extended
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literature scan enabled the identification of constraints originating from diverse educational systems and modelling traditions, thereby enriching the catalogue beyond the local institutional context and improving its general applicability.

3.1.3 Constructed Constraints

To enhance the completeness and usability of the constraint catalogue, we introduced a set of constructed constraints derived from simple transformations or generalizations of existing ones. These constraints were not explicitly identified in either the institutional survey or the literature review, but they logically follow from common patterns and scheduling logic.

For instance, the constraint “Exam A shall be scheduled before Exam B”, observed at Université Catholique de l’Ouest ([Boizumault et al. \(1996\)](#)), was complemented by its inverse “Exam A shall be scheduled after Exam B”, which, while not directly reported, is a trivial and practically relevant counterpart.

Constraints constructed in this way are explicitly labeled as “constructed constraint” in the catalogue, to distinguish them from those directly sourced from institutional or academic references.

3.2 Constraint Categorization and indexing

In many studies, constraints are classified into two main types: hard constraints, whose violation renders a timetable infeasible, and soft constraints, whose violation only reduces the quality or objective score of a solution. However, this classification is not universal. Its interpretation depends heavily on the institutional context. At a minimum, each institution must treat its legal and regulatory obligations as hard constraints yet such obligations vary considerably across contexts.

Accordingly, our catalogue is organized using a hierarchical classification system:

- At the top level, constraints are grouped by broad functional domains.

231 - At the second level, each domain is subdivided into specific subcategories, each
232 capturing a particular set of scheduling rules.
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234 Within each subcategory, we list variants of constraints drawn from different insti-
235 tutions and publications, in order to reflect the diversity of how similar issues are
236 handled across contexts.
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238 Some constraints may be relevant to more than one category. In such cases, they
239 are listed under multiple headings and explicitly marked as equivalent constraints to
240 indicate their conceptual similarity.
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242 Below is an overview of the main top-level categories used in the catalogue:
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244 **3.2.1 Room and Facility**

245 This category includes constraints related to the physical characteristics of exam-
246 ination venues: room capacity, required equipment, environmental conditions (e.g.,
247 lighting or accessibility), spatial preferences, and room-sharing limitations.
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249 **3.2.2 Student Conflict and Spread**

250 This group covers both conflict-avoidance constraints (e.g., a student cannot be
251 scheduled for overlapping exams) and spread constraints that aim to reduce stu-
252 dent workload over short time windows (e.g., “no more than one exam per day” or
253 “minimum two-hour gap between exams”). These rules promote equity and student
254 well-being, which are critical for timetable acceptance.
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256 **3.2.3 Invigilation and Staffing**

257 These constraints govern the human resources involved in exam supervision. They
258 include minimum staffing levels per room, invigilator availability, required skills (e.g.,
259 language proficiency), rest periods, and fair workload distribution. Integrating these
260 constraints early in the scheduling process helps avoid conflicts with institutional HR
261 policies.
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3.2.4 Sequencing and Dependencies	277
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Sequencing constraints define the required order between exams whether based on	279
pedagogical logic (e.g., prerequisites) or structural dependencies (e.g., joint assessment	280
of lab and theory components). These constraints preserve curricular coherence and	281
prevent logical inconsistencies in exam ordering.	282
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3.2.5 Equity and Inclusion	287
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This category includes constraints ensuring accommodations for students with disabil-	289
ities, religious observances, or linguistic requirements. Encoding these needs explicitly	290
not only ensures compliance with legal frameworks but also reinforces institutional	291
commitments to inclusion and diversity.	292
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3.2.6 Logistics and Cost	297
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Finally, this category groups constraints aimed at improving operational efficiency.	299
Examples include avoiding exam sessions during public transport peak hours, min-	300
imizing building use after hours, or reducing the use of costly resources. These are	301
typically modeled as soft constraints and provide decision-makers with quantitative	302
levers to balance cost, comfort, and accessibility.	303
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4 Results	309
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Our survey resulted in a well-balanced set of 117 constraints , distributed across the	311
following categories:	312
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Europe, North America, and the Middle East/Gulf regions are well represented in	314
the sample. However, no institutions from Oceania or South America were identified,	315
and only one African institution is included. Several major higher-education systems	316
such as those of Japan, China, Brazil, India, Pakistan, Mexico, Germany, and Russia	317
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Table 1 Distribution of constraints by category

Category	Number of Constraints
Room & Facility	22
Student Conflict & Spread	15
Invigilation & Staffing	20
Sequencing & Dependencies	19
Equity & Inclusion	17
Logistics & Cost	25
Total	117

are also absent. This highlights a clear avenue for future data collection and expansion of the catalogue.

Table 2 Distribution by country

Country	Institutions	Constraints
France	3	38
United Kingdom	2	21
Multi-country / Unspecified	7	17
Malaysia	3	12
Turkey	3	7
United States	3	7
Kuwait	1	4
Saudi Arabia	2	4
Belgium	1	3
Canada	2	3
Greece	1	3
United Arab Emirates	2	2

Table 3 Distribution by continent

Continent	Institutions	Constraints
Europe	15	66
Middle East & Gulf	8	18
Asia-Pacific	7	12
North America	6	10
Africa	1	1
Multi-region / Unspecified	1	17

The sample spans institutions of varying sizes and governance models, including public, private, faith-based, and military institutions, thereby providing a diverse foundation for future benchmarking efforts.

Table 4 Distribution by institutional status

Status Category	Institutions	Constraints
Public	17	71
Private	4	4
Confessional / Faith-based	2	6
Military	1	5

Table 5 Distribution by institution size

Size Band	Institutions	Constraints
Large	9	23
Medium (< 30,000 students)	9	46
Small (< 5,000 students)	6	17
Unspecified	9	38

Some constraints listed in the catalogue may no longer be applicable, while others may apply only to specific faculties within an institution. When a source indicated a constraint already present in the catalogue, it was not duplicated in order to limit the size of the appendices. In addition, a small number of constraints were introduced by the authors despite not being explicitly stated in the sources; these were derived from existing constraints for completeness and are explicitly marked as constructed constraints.

For readability, the complete catalogue of constraints is provided in Appendix B to G and at <https://github.com/Kaputchino/Constraints-for-the-Examination-Timetabling-Problem>.

5 Conclusion

This study presents a comprehensive catalogue of 117 exam timetabling constraints, derived from fieldwork, academic literature, and systematic construction. While many constraints are widely shared across institutions, others are shaped by local pedagogical, legal, or resource-specific factors. The hierarchical structure of the catalogue by domain and subcategory offers a modular framework that facilitates adoption, adaptation, and comparison across settings.

The geographic coverage, although diverse in governance models and institution sizes, remains incomplete. Notably, no institutions from Oceania or South America were included, and large systems such as Japan, China, India, Germany, and Brazil

are absent. Incorporating constraints from these regions would significantly enhance
the global relevance of the catalogue.

To increase the practical value of this work, future directions could include identifying how best to select or prioritize constraints based on institutional context, one envisaged method is the wisdom of the crowd.

Appendix A Journals

Publisher	Journal
Elsevier	Applied Soft Computing
	European Journal of Operational Research
	Technological Forecasting and Social Change
	Operations Research Letters
	Surveys in Operations Research and Management Science
INFORMS	Operations Research Perspectives
	Mathematics of Operations Research
	Information Systems Research
	Organization Science
	INFORMS Journal on Computing
	Service Science
	Operations Research
Springer	Management Science
	Manufacturing and Service Operations Management
	Central European Journal of Operations Research
	Operations Research / Computer Science Interfaces Series
	Operations Management Research
	Mathematical Methods of Operations Research
	International Series in Operations Research and Management Science
	Operations Research Forum
	Journal of the Operations Research Society of China
	Annals of Operations Research
Taylor & Francis	4OR
	Journal of Mathematical Modelling and Algorithms in Operations Research
Wiley	Operational Research
	Journal of the Operational Research Society
EDP Sciences	Journal of Information Technology Case and Application Research
Operations Research Society of Japan	International Transactions in Operational Research
Croatian Operational Research Society	Advances in Operations Research
University of Belgrade	RAIRO – Operations Research
University of the Punjab	Journal of the Operations Research Society of Japan
Wrocław University of Science and Technology	Croatian Operational Research Review
	Yugoslav Journal of Operations Research
	Pakistan Journal of Statistics and Operation Research
	Operations Research and Decisions

Table A1 Non-exhaustive list of journals in related fields by publisher

Appendix B Room & facility

Constraint	Variant	Source
R1 Capacity	R1-1 The seating capacity of the assigned room shall not be lower than the number of registered candidates.	Université de Tours
	R1-2 The student-to-seat ratio for any examination shall not exceed a pre-defined limit.	Université de Tours
	R1-3 The total number of candidates examined on a given calendar day shall not exceed an institution-specific ceiling.	Cornell University (Ye et al. (2025))
	R1-4 The aggregate number of students present within a single building at any time shall not surpass a capacity limit set by safety regulations.	Université de Tours
	R1-5 For oral examinations, the number of examinees in a time slot shall not exceed the maximum interview capacity authorised by the department.	Katholieke Hogeschool (Demeester et al. (2012))
	R1-6 The number of distinct examinations assigned to the same room and time slot shall not exceed a predefined maximum N.	Hubei University of Technology (Ou Yang and Chen (2010))
	R1-7 No more than 1200 students to sit an exam at any one time.	Swansea University (Thompson and Dowsland (1995))
	R2-1 A buffer of not less than a specified time interval shall separate consecutive examinations scheduled in the same room.	Université de Tours
	R2-2 The temperature shall remain within an acceptable range, or the room must offer appropriate climate control.	Université de Tours
	R2-3 The room shall remain continuously available for the entire duration of the examination.	Université de Tours
R3 Location / Preference	R3-1 Preferentially assign examinations to rooms located within the candidates' departmental building.	Israel Institute of Technology (Strichman (2017))
	R3-2 Certain examinations shall be scheduled exclusively in designated rooms.	Université de Tours
	R3-3 Large exam halls must be scheduled in preference to smaller ones.	Unknown British University (Burke et al. (1996))

599	Constraint	Variant	Source
600		R3-4 Exams must be scheduled to rooms near to the relevant department.	Unknown British University (Burke et al. (1996))
601		R3-5 Exams for the same department should be kept together.	Unknown British University (Burke et al. (1996))
602		R3-6 Specific rooms may be reserved exclusively for particular degree programmes.	Yıldız Technical University (Güler and Geçi'Ci' (2020))
603		R3-7 Student which has consecutive exams on the same day should be assigned to the same room.	Universiti Kebangsaan Malaysia (Ayob et al. (2007))
604		R3-8 A penalty is incurred when examinations for distinct courses are placed in the same building or in adjacent rooms.	University of Sharjah (Lulu et al. (2024))
605		R3-9 Rooms allocated to a split examination shall, where possible, be located in the same building or campus sector.	University Malaysia Sarawak (Siew et al. (2025))
606	R4 Merger	R4-1 The examination shall not be merged with any other examination.	Université de technologie de Compiègne (Arbaoui et al. (2013))
607		R4-2 The examination may be merged with other examinations.	Université de Tours
608		R4-3 Only examinations belonging to authorised categories may be merged.	Universiti Kebangsaan Malaysia (Ayob et al. (2007))
609		R4-4 Oral and written components shall not be merged into the same room.	Katholieke Hogeschool Sint-Lieven (Demeester et al. (2012))
610		R4-5 Candidates shall be split into separate rooms according to examination end times.	Unknown British University (Burke et al. (1996))
611		R4-6 Examinations of differing duration shall not be scheduled in the same room.	Unknown British University (Burke et al. (1996))
612		R4-7 The number of rooms assigned to a split examination shall be minimised.	University Malaysia Sarawak (Siew et al. (2025))
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Appendix C Student Conflict & Spread

Category	Variant	Source
S1 Load	S1-1 A minimum gap of hours shall separate any two examinations taken by the same student on the same day.	Université de Tours
	S1-2 Subjects designated as difficult shall be placed toward the end of the examination session to mitigate early cognitive overload.	Komijan and Koupaei (2012)
	S1-3 Each student's examinations should be evenly distributed across the timetable.	Université Catholique de l'Ontario (Boizumault et al. (1996))
	S1-4 No student shall be required to sit more than a set number of examinations per day.	Université de Sfax (Dammak et al. (2017))
	S1-5 For a given academic programme, the number of examinations per day shall not exceed the programme-specific daily limit.	Ankara HBV University (Bayar and Uzun Bayar (2020))
	S1-6 The cumulative examination duration assigned to any student over any two consecutive days shall not exceed the maximum total duration set by policy.	University College Cork (Genc and O'Sullivan (2020))
	S1-7 Within a single academic department, the number of examinations scheduled on the same day shall not exceed the departmental daily limit.	Yıldız Technical University (Güler et al. (2021))
	S1-8 If a student has only one examination on a given day, that examination should be placed outside the evening time slots to reduce on-campus idle time.	Kamisli Ozturk et al. (2024)
	S1-9 The number of consecutive examinations assigned to a student within the same day shall not exceed the permitted consecutive-exam limit.	Unknown British University (Burke et al. (1996))
	S1-10 Take-away examinations should not be scheduled concurrently with in-person examinations.	Kamisli Ozturk et al. (2024)
	S1-11 A student shall not be assigned an examination in the last time slot of one day followed by another in the first slot of the next day, to ensure sufficient overnight rest.	Middle East Technical University (Ergül (1996))
	S1-12 The average difficulty of a student's examinations shall be fairly distributed.	Muklason et al. (2017)
	S1-13 A penalty shall be applied to examinations scheduled on weekends.	MARA University of Technology (Kendall and Hussin (2005))

Category	Variant	Source
	S1-14 All examinations belonging to the same academic year shall be scheduled in a single designated time band (morning, afternoon, or evening).	University of Toronto (Carter et al. (1994))
	S1-15 A scheduling penalty shall be applied when a student is assigned examinations in both the period before lunch and after lunch on the same day.	Di Gaspero and Schaerf (2001)
	S1-16 Examinations for courses in the same programme and term shall not be scheduled in consecutive time slots on the same day.	Universidad Diego Portales (Cataldo et al. (2017))
S2 Overlap	S2-1 A student shall not be allocated to two examinations occurring in the same time slot.	Université de Tours
	S2-2 Re-sit candidates must be prevented from overlapping with another examination.	Université de Tours
	S2-3 Examination clashes for each academic programme should be minimised.	Ankara HBV University (Bayar and Uzun Bayar (2020))
	S2-4 Examination clashes for each programme-year combination should be minimised.	Ankara HBV University (Bayar and Uzun Bayar (2020))
	S2-5 Scheduling should minimise clashes among elective courses.	Ankara HBV University (Bayar and Uzun Bayar (2020))
	S2-6 Exams A and B must not share the same period. (Equivalent to D1-4)	United States Military Academy in West Point (Wang et al. (2010))
	S2-7 Examinations for courses belonging to the same academic programme and term shall not share the same time slot.	Universidad Diego Portales (Cataldo et al. (2017))
	S2-8 Minimise the disparity in exam-slot conflicts among academic programmes and terms.	Universidad Diego Portales (Cataldo et al. (2017))
S3 Assignment	S3-1 Minimise the number of students assigned to examination commissions where neither their supervisor nor referee serves as examiner.	Masaryk University (Kochaniková and Rudová (2013))
	S3-2 Students supervised (or reviewed) by the same faculty member shall be scheduled in consecutive examination sequences so that the faculty member attends the minimum possible number of separate sequences. Equivalent to I1-10	Masaryk University (Rudová et al. (2014))
	S3-3 The thesis should be scheduled such as the supervisor and the referee of the thesis examination can both attend. Equivalent to I3-6	Masaryk University (Kochaniková and Rudová (2013))

Appendix D Invigilation & Staffing

Category	Variant	Source
I1 Workload	I1-1 The cumulative invigilation workload assigned to any staff member shall not exceed the maximum load defined by institutional policy.	King Saud University (Hosny (2012))
	I1-2 Invigilation duties shall not be assigned in the late-evening time band more frequently than the frequency limit specified by the institution.	King Saud University (Al-Neghemish et al. (2018))
	I1-3 Each instructor shall have at least one unscheduled day per examination week.	King Saud University (Al-Neghemish et al. (2018))
	I1-4 Staff holding additional roles (e.g., administration, funded research) shall receive a reduced number of invigilation duties.	University Technology MARA (Cowling et al. (2002))
	I1-5 The number of examinations invigilated by any staff member during the exam period shall not exceed the staff-duty limit specified by the institution.	Universiti Malaysia Pahang (Mohamad Kahar and Kendall (2014))
	I1-6 Invigilation duties shall be balanced among faculty members holding the same academic rank.	King Saud University (Al-Neghemish et al. (2018))
	I1-7 Chief-invigilator duties shall be evenly spread among eligible lecturers.	Universiti Malaysia Pahang (Mohamad Kahar and Kendall (2014))
	I1-8 Invigilation assignments shall be distributed so that each invigilator carries a comparable total duty load.	University Technology MARA (Cowling et al. (2002))
	I1-9 A minimum rest period of hours shall separate consecutive invigilation duties assigned to the same staff member.	University Technology MARA (Cowling et al. (2002))
	I1-10 Students supervised (or reviewed) by the same faculty member shall be scheduled in consecutive examination sequences so that the faculty member attends the minimum possible number of separate sequences. Equivalent to S3-2.	Masaryk University (Rudová et al. (2014))
	I1-11 Minimise the number of proctor schedules containing split shifts on the same day.	Carleton University (Awad and Chinneck (1998))
	I1-12 Avoid evening-to-morning schedules.	Carleton University (Awad and Chinneck (1998))
	I1-13 Distribute evening-to-morning schedules evenly among proctors.	Carleton University (Awad and Chinneck (1998))

783	Category	Variant	Source
784		I1-14 Allocate a greater number of invigilation shifts to proctors in higher experience classes.	Carleton University (Awad and Chinneck (1998))
785		I1-15 Within the same experience class, assign more shifts to proctors who are available for more shifts and willing to work in more rooms.	Carleton University (Awad and Chinneck (1998))
786		I1-16 Assign fewer unfavourable schedules to proctors in higher experience classes.	Carleton University (Awad and Chinneck (1998))
787	I2 Roles / Separation	I2-1 An instructor shall not invigilate an examination for which they are the principal teacher.	University Technology MARA (Cowling et al. (2002))
788		I2-2 Teaching staff shall not simultaneously hold the roles of teacher and invigilator for the same examination.	Ghaffar et al. (2025)
789		I2-3 Where pedagogical policy requires it, the same individual may serve as teacher, subject expert, supervisor, and invigilator for a designated examination.	Ghaffar et al. (2025)
790		I2-4 Of the professors assigned to a session, at least one shall be the principal examiner and at least one an external supervisor.	Vestfold County School Department (Avela et al. (2025))
791		I2-5 There should be one instructor at each timeslot who can be the chair.	Budapest University of Technology and Economics (Erdős and Kovári (2021))
792		I2-6 There should be one instructor at each timeslot who can be the secretary.	Budapest University of Technology and Economics (Erdős and Kovári (2021))
793		I2-7 There should be at least one instructor at each timeslot who can be the member.	Budapest University of Technology and Economics (Erdős and Kovári (2021))
794		I3-1 An invigilator shall be available for the entire duration of the examination to which they are assigned.	Université de Tours
795	I3 Availability	I3-2 A minimum buffer of hours shall separate an instructor's own examination from any subsequent invigilation duty.	University Technology MARA (Cowling et al. (2002))
796		I3-3 Staff involved in project presentations or evaluations shall not be scheduled for invigilation during the conflicting time blocks.	University Technology MARA (Cowling et al. (2002))
797		I3-4 A guard-time parameter shall be observed before an invigilator may supervise a second examination.	Kuwait University (Al-Yakoob et al. (2010))
798		I3-5 Instructor time-slot preferences should be honoured where feasible.	Université de Tours
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Category	Variant	Source
	I3-6 The supervisor and the referee of a thesis examination must be scheduled so that both can attend. Equivalent to S3-3.	Masaryk University (Kochaniková and Rudová (2013))
	I3-7 Examinations for courses taught by the same instructional team shall be scheduled in different time slots.	Universidad Diego Portales (Carrado et al. (2017))
	I3-8 Car-pool members always work the same shifts.	Carleton University (Awad and Chinneck (1998))
I4 Requirement	I4-1 Each examination shall be supervised by at least the minimum number of invigilators specified by regulation.	Université de Tours
	I4-2 Each academic department shall provide at least the departmental quota of proctoring personnel per examination day.	Ankara HBV University (Uzun Bayar and Uzun Bayar (2020))
	I4-3 Designated examinations shall be supervised by specifically named invigilators.	Université de Tours
	I4-4 Where required, female proctors shall be assigned to examinations held for female-only cohorts. (Equivalent to E2-2)	University of Kuwait (Al-Yaqub et al. (2010))
	I4-5 Each campus shall supply invigilators for its own examinations using its assigned pool of teaching staff.	University Technology MARA (Kaplansky et al. (2004))
	I4-6 Each shift shall include any full-time proctor who is available for that shift.	Carleton University (Awad and Chinneck (1998))
	I4-7 A minimum of two professors qualified for the subject shall be assigned to every active examination session.	Vestfold County School Department (Avella et al. (2025))

Appendix E Sequencing & Dependencies

Category	Variant	Source
D1 Sequence	D1-1 Exam A shall be scheduled before exam B.	Université Catholique de l'Ouest (Boizumault et al. (1996))
	D1-2 Exam A shall be scheduled after exam B.	constructed constraint
	D1-3 Exams A and B shall be scheduled in the same period.	Université de technologie de Compiègne (Arbaoui et al. (2013))
	D1-4 Exams A and B must not share the same period. (Equivalent to S2-6)	United States Military Academy in West Point (Wang et al. (2010))
	D1-5 Exams sharing common questions shall be scheduled concurrently.	Unknown British University (Burke et al. (1996))
	D1-6 Exam A must precede exam B by at least a pre-defined time interval .	Université de Tours
	D1-7 Exam A must follow exam B by at least a pre-defined time interval .	Université de Tours
	D1-8 Exactly one exam in a designated group shall be scheduled last within that group.	Unknown British University (Burke et al. (1996))
	D1-9 Exactly one exam in a designated group shall be scheduled first within that group.	Unknown British University (Burke et al. (1996))
	D1-10 Specified pairs of exams shall be held on the same calendar day.	University of Toronto (Carter and Laporte (1996))
	D1-11 Exams belonging to mutually exclusive categories (e.g., two music classes) shall not be scheduled simultaneously.	Cedar Crest College (Mehta (1981))
	D1-12 Where feasible, avoid scheduling designated courses in specified examination periods.	United States Military Academy in West Point (Wang et al. (2010))
	D1-13 All written papers of the same subject shall be scheduled in a single period.	Katholieke Hogeschool Sint-Lieven (Demeester et al. (2012))
	D1-14 Exams E and E shall be scheduled in the same room in the same time.	University of British Columbia (Hare and Hamilton (2022))
D2 Fixed Time	D2-1 An examination should be held at the same time as the regular lecture where institutional policy so requires.	École de technologie supérieure (Wong et al. (2002))
	D2-2 An examination must be scheduled no later than a pre-defined deadline.	constructed constraint
	D2-3 An examination must be scheduled no earlier than a pre-defined release time.	constructed constraint

Category	Variant	Source	921
	D2-4 An examination must be held at a fixed, pre-assigned period.	Université Catholique de l'Ouest (Boizumault et al. (1996))	922 923
	D2-5 The final examination for a course shall be scheduled only on weekdays that match its regular meeting pattern.	Kutztown University (Sienko et al. (2016))	924 925 926 927
D3 Dependent Scheduling	D3-1 Examinations with the largest candidate counts should be scheduled early in the timetable.	Unknown British University (Bozke et al. (1996))	928 929
	D3-2 High-capacity courses should be timetabled in the first week to maximise grading time for lecturers.	Kamisli Ozturk et al. (2024)	930 931 932
	D3-3 Finals, honours examinations, or papers subject to external moderation should appear early in the session.	Unknown British University (Bozke et al. (1996))	933 934
	D3-4 Examinations with the largest candidate counts shall be scheduled toward the end of the examination session.	Alzaqebah and Abdullah (2015)	935 936 937 938
	D3-5 Designated examinations shall rotate their position within the timetable across successive academic years so that no single exam is always last.	University of Toronto (Carter et al. (1994))	939 940 941
	D3-6 Retain the optimised timetable from the previous year as much as possible (minimise changes).	Carter and Laporte (1996)	942 943 944
	D3-7 Examinations shall be fixed in hierarchical order.	King Fahd University of Petroleum and Minerals (Almaslami et al. (2023))	945 946

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Appendix F Equity & Inclusion

Category	Variant	Source
E1 Handicap	E1-1 Authorised students with disabilities shall receive extra time in accordance with accessibility policy.	Université de Tours
	E1-2 Students with disabilities shall be scheduled in rooms that meet accessibility standards (ramps, adapted desks, etc.).	Université de Tours
	E1-3 Required assistive equipment (e.g., screen readers, enlarged-font papers) shall be available at the examination venue.	Université de Tours
	E1-4 Dedicated support staff shall be assigned to assist candidates with disabilities during examinations.	Université de Tours
	E1-5 Accessible examination rooms shall be reserved for invigilators with disabilities.	Université de Tours
	E1-6 Students with disabilities shall be scheduled in separated rooms.	Université de Tours
E2 Religion / Culture	E2-1 Religious convictions must be respected.	Unknown British University (Burke et al. (1996))
	E2-2 Where required, female proctors shall be assigned to examinations held for female-only cohorts. (Equivalent to I4-4)	University of Kuwait (Al-Yakoob et al. (2010))
	E2-3 If an examination falls on a state-specific public holiday observed by any registered candidate, the examination shall be moved to an alternative slot. (Equivalent to L1-7)	University Technology MARA (Cowling et al. (2002))
E3 Limited Time	E3-1 The scheduling algorithm shall respect the declared availability windows of students on work–study (alternance) programmes.	Unknown British University (Burke et al. (1996))
	E3-2 The availability constraints of part-time students shall be honoured when allocating examination periods.	Unknown British University (Burke et al. (1996))
	E3-3 Designated student groups may be restricted to a predefined subset of examination periods.	University of Toronto (Carter and Laporte (1996))
E4 Language	E4-1 Foreign-exchange students shall, where feasible, be seated in separate rooms from domestic cohorts.	Unknown British University (Burke et al. (1996))
	E4-2 The assigned invigilator shall be proficient in the language of the examination.	École des Mines de Nantes (David (1998))

Category	Variant	Source	1013
	E4-3 Where practicable, examinations delivered in different languages should not occur on the same day for the same cohort of students.	Unknown British University (Burke et al. (1996))	1014 1015 1016 1017
E5 Diversity	E5-1 Certain examination rules may apply only to specified categories of classes or students (e.g., clinical placements).	United States Military Academy in West Point (Wang et al. (2010))	1018 1019 1020
	E5-2 Re-sit candidates should be scheduled in locations distinct from first-sit candidates.	Unknown British University (Burke et al. (1996))	1021 1022 1023
	E5-3 Students enrolled in adjacent-year curricula (AJAC) shall be treated as a distinct cohort for clash-avoidance purposes.	Université de Tours	1024 1025 1026

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Appendix G Logistics & Cost

Category	Variant	Source
L1 Window & Session	L1-1 All examinations shall be scheduled within the official examination window published by the university.	Université de Tours
	L1-2 Designated examinations may be timetabled only inside a predefined subset of periods.	Unknown British University (Burke et al. (1996))
	L1-3 The length of the overall examination session shall be minimised with respect to the number of calendar days required.	Athens University of Economics and Business (Dimopoulou and Miliotis (2001))
	L1-4 The total number of examination days shall be minimised subject to feasibility.	Masdar Institute (Siew et al. (2024))
	L1-5 Each examination period shall have a minimum duration prescribed by institutional policy.	Université de Tours
	L1-6 Every examination day shall consist of a set of predefined time periods.	Athens University of Economics and Business (Dimopoulou and Miliotis (2001))
	L1-7 If an examination falls on a state-specific public holiday observed by any registered candidate, the examination shall be moved to an alternative slot. (Equivalent to E2-3)	University Technology MARA (Cowling et al. (2002))
	L1-8 Any free examination periods shall be placed at the end of the timetable.	École des Mines de Nantes (David (1998))
	L1-9 Maximise the number of exams that could be moved to another period without disrupting the rest of the schedule.	Carter and Laporte (1996)
L2 Timing & Buffers	L2-1 Timetables shall allow sufficient travel time for students moving between distinct campuses or sites.	Unknown British University (Burke et al. (1996))
	L2-2 An examination shall not be split across multiple periods.	Université de technologie de Compiègne (Arbaoui et al. (2013))
	L2-3 The duration of an examination assigned to a period shall not exceed the duration of that period.	Université de technologie de Compiègne (Arbaoui et al. (2013))
	L2-4 A minimum transfer buffer shall separate two examinations when a candidate moves between examination centres. (Equivalent to L5-5)	Modirghorasani and Hoseinpour (2024)
L3 Location	L3-1 Students should, where practicable, be examined on their home site.	Unknown British University (Burke et al. (1996))
	L3-2 Consecutive examinations for the same student should not be held at different sites.	Unknown British University (Burke et al. (1996))

Category	Variant	Source	1105
	L3-3 Each candidate shall be assigned to a specific examination centre. (Equivalent to L5-4)	Modirghorasani and Hoseinpour (2024)	1106 1107
	L3-4 Scheduling shall account for parking capacity and expected traffic conditions around examination venues.	Kuwait University (Al-Yakoob et al. (2010))	1108 1109
	L3-5 Each student shall be assigned to the geographically nearest examination centre, minimising travel distance.	Çalik et al. (2024)	1110 1111 1112 1113
	L3-6 Proctors shall not be assigned to rooms they have flagged as unacceptable or unadapted.	Carleton University (Awad and Chinneck (1998))	1114 1115 1116
L4 Cost	L4-1 Usage of premium-cost resources (e.g., rented halls) shall be penalised in the objective function.	Université de Tours	1117 1118
	L4-2 Rooms with higher rental rates shall incur a cost penalty proportional to their use.	Université de Tours	1119 1120
	L4-3 Hiring temporary invigilation staff shall be modelled with a unit cost to be minimised.	Université de Tours	1121 1122 1123
L5 Pandemic	L5-1 To mitigate pandemic risks, examinations should be assigned to department-specific time slots, limiting inter-department contact.	Kamisli Ozturk et al. (2024)	1124 1125 1126
	L5-2 High-capacity examinations should avoid early-morning slots to reduce congestion in public transport and lower transmission risk.	Kamisli Ozturk et al. (2024)	1127 1128 1129 1130
	L5-3 Candidates may be separated in different centres to limit pandemic risk.	Modirghorasani and Hoseinpour (2024)	1131 1132
	L5-4 Each candidate shall be assigned to a specific examination centre. (Equivalent to L3-3)	Modirghorasani and Hoseinpour (2024)	1133 1134
	L5-5 A minimum transfer buffer shall separate two examinations when a candidate moves between centres. (Equivalent to L2-4)	Modirghorasani and Hoseinpour (2024)	1135 1136 1137 1138
			1139
L6 Repetition	L6-1 Every examination must be scheduled.	Athens University of Economics and Business (Dimopoulou and Miliotis (2004))	1140 1141
	L6-2 Every examination must be scheduled at least once.	United States Military Academy in West Point (Wang et al. (2010))	1142 1143
	L6-3 Every examination must be scheduled only once.	Université de Tours	1144

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